

Problems in Correlation

1)

Compute the coefficient of correlation between X and Y, using the following data:

X	:	1	3	5	7	8	10
Y	:	8	12	15	17	18	20

Ans: 0.9879

2)

Compute the coefficients of correlation between X and Y using the following data:

X	:	65	67	66	71	67	70	68	69
Y	:	67	68	68	70	64	67	72	70

Ans: 0.3974

3)

Find the coefficient of correlation between X and Y using the following data:

X	:	5	10	15	20	25
Y	:	16	19	23	26	30

Ans: 0.9907

4)

The following table gives the bivariate frequency distribution of marks in an intelligence test obtained by 100 students according to their age:

Age (x) in yrs Marks (y)	18	19	20	21	Total
10–20	4	2	2	–	8
20–30	5	4	6	4	19
30–40	6	8	10	11	35
40–50	4	4	6	8	22
50–60	–	2	4	4	10
60–70	–	2	3	1	6
Total	19	22	31	28	100

Calculate the coefficient of correlation between age and intelligence.
Ans: 0.1897

5)

Calculate the correlation coefficient for the following ages of husbands (X) and wives (Y), using only standard deviations of X and Y :

X:	23	27	28	28	29	30	31	33	35	36
Y:	18	20	22	27	21	29	27	29	28	29

Ans: 0.8176

6)

If the independent random variables X and Y have the variances 36 and 16 respectively, find the correlation coefficient between $(X + Y)$ and $(X - Y)$.

Ans: 5/13

7)

If X, Y and Z are uncorrelated RV's with zero means and standard deviations 5, 12 and 9 respectively and if $U = X + Y$ and $V = Y + Z$, find the correlation coefficient between U and V.

Ans: 48/65

8)

If X and Y are two RV's with variances σ_X^2 and σ_Y^2 respectively, find the value of k, if $U = X + kY$ and $V = X + \frac{\sigma_X}{\sigma_Y} \cdot Y$ are uncorrelated.

9)

If (X, Y) is a two-dimensional RV uniformly distributed over the triangular region R bounded by $y = 0$, $x = 3$ and $y = 4/3 x$. Find $f_X(x)$, $f_Y(y)$, $E(X)$, $\text{Var}(X)$, $E(Y)$, $\text{Var}(Y)$ and ρ_{XY} .